

Weil Conjectures for Elliptic Curves

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Table des matières

- 1 Motivation
- 2 Vocabularies
- 3 Towards the Question
- 4 Differential
- 5 Riemann-Roch
- 6 Tate Module

Riemann Hypothesis

Conjecture (Riemann Hypothesis)

The Riemann zeta function is defined as

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

Then, the non-trivial zeros of the Riemann zeta function all have a real part equal to $1/2$.

Zeta Function for Projective Varieties

Definition (Zeta Function for Projective Varieties)

Let V be a projective variety over $\mathbb{F}_q$. The zeta function of V is the formal series

$$Z(V/\mathbb{F}_q; T) = \exp \left(\sum_{n=1}^{\infty} (\# V(\mathbb{F}_{q^n})) \frac{T^n}{n} \right) \in \mathbb{Q}[[T]],$$

where $\exp(F(T))$ denotes the formal series $\sum_{i=0}^{\infty} F(T)^i / i!$ for a formal series without a constant term.

Weil Conjecture

Taking a change of variable $T = q^{-s}$, we define the zeta function as:

$$\zeta(s) = Z(E/\mathbb{F}_q; q^{-s}).$$

Theorem (Weil Conjecture (Part))

(Deligne 1973) Let V be a smooth projective variety over $\mathbb{F}_q$ of dimension n . The Riemann hypothesis holds for the zeta function. That is, the zeros of the zeta function all have a real part equal to $1/2$.

Main Theorem

GOAL: Weil Conjectures for Elliptic Curves, in other words:

Theorem (Main Theorem)

(Hasse) Let E be an elliptic curve over $\mathbb{F}_q$. Then there exists $a \in \mathbb{Z}$ such that

$$Z(E/\mathbb{F}_q; T) = \frac{(1 - \alpha T)(1 - \beta T)}{(1 - T)(1 - qT)} \text{ (Rationality),}$$

$$Z(E/\mathbb{F}_q; \frac{1}{qT}) = Z(E/\mathbb{F}_q; T) \text{ (Functional Equation),}$$

and

$$1 - aT + qT^2 = (1 - \alpha T)(1 - \beta T) \text{ with } |\alpha| = |\beta| = \sqrt{q} \text{ (Riemann Hypothesis).}$$

Table des matières

- 1 Motivation
- 2 Vocabularies
- 3 Towards the Question
- 4 Differential
- 5 Riemann-Roch
- 6 Tate Module

Projective Curve

Definition (Projective Curve)

Let the projective plane $\mathbb{P}^2(k)$ be the quotient $(k^3 \setminus \{0\})/k^\times$. The class of a point $\tilde{P} = (x_0, x_1, x_2) \in \mathbb{A}^3$ is denoted $P := [x_0, x_1, x_2]$ in $\mathbb{P}^2(k)$. A projective curve is a closed subvariety of a projective plane. It can be written as

$$V(f) := \{[x_0, x_1, x_2] \in \mathbb{P}^2 : f([x_0, x_1, x_2]) = 0\}$$

where f is a homogeneous polynomial in $k[T_0, T_1, T_2]$.

Why projective?

Example (Weierstrass Equation)

$$Y^2Z + a_1XYZ + a_3YZ^2 = X^3 + a_2X^2Z + a_4XZ^2 + a_6Z^3.$$

Anti-Equivalence

Theorem

There is an anti-equivalence between the two categories:

$$\left[\begin{array}{ll} \text{Objects:} & \text{extensions of } k \\ & \text{of finite type and} \\ & \text{of transcendence} \\ & \text{degree } 1 \\ \text{Morphisms:} & \text{field extensions} \end{array} \right] \rightleftarrows \left[\begin{array}{ll} \text{Objects:} & \text{smooth} \\ & \text{projective} \\ & \text{curves} \\ \text{Morphisms:} & \text{dominant} \\ & \text{morphisms} \end{array} \right]$$

Anti-Equivalence

Démonstration.

(Idea)

- $\rightsquigarrow$: Just take the function field.
- $\leftarrow$: for K/k , a finite type extension of transcendence degree 1, suppose

$$C_K := \{\text{valuation of } K/k \text{ } v : K^\times \twoheadrightarrow \mathbb{Z}\}$$

endowed with the cofinite topology. We can give a sheaf structure on C_K and verify that it is indeed a smooth projective curve.



Elliptic Curve

Definition (Elliptic Curve)

An elliptic curve over k is a pair (E, O) where E is a smooth projective curve of genus 1 defined over k and O is a k -rational point.

Degree

Definition (Degree, Algebraic Side)

Let $\varphi : C \rightarrow C'$ be a dominant morphism. We define its degree $\deg(\varphi) := [k(C) : k(C')]$, and its separable degree $\deg_s(\varphi) := [k(C)_{\text{sep}} : k(C')] = [k(C) : k(C')]_s$.

Lemma (Geometric Side)

Let $\varphi : C \rightarrow C'$ be a morphism between two smooth projective curves, then for any $Q \in C'$,

$$\deg(\varphi) = \sum_{P \in \varphi^{-1}(Q)} e_P(\varphi)$$

Table des matières

- 1 Motivation
- 2 Vocabularies
- 3 Towards the Question
- 4 Differential
- 5 Riemann-Roch
- 6 Tate Module

Proposition

Proposition

Let $\mathbb{F}_{q^n}$ denote the extension of $\mathbb{F}_q$ of degree n , and $E(\mathbb{F}_{q^n})$ the $\mathbb{F}_{q^n}$ -rational points of E . We have

$$\#E(\mathbb{F}_{q^n}) = 1 - \alpha^n - \beta^n + q^n$$

where $\alpha, \beta \in \mathbb{C}$ are two complex conjugate numbers with ordinary norm $\sqrt{q}$ in $\mathbb{C}$.

Proof of the Main Theorem Assuming the Previous Proposition.

$$\begin{aligned}\log Z(E/\mathbb{F}_q; T) &= \sum_{n=1}^{\infty} (1 - \alpha^n - \beta^n + q^n) \frac{T^n}{n} \\ &= -\log(1 - T) + \log(1 - \alpha T) + \log(1 - \beta T) \\ &\quad - \log(1 - qT).\end{aligned}$$

This implies that

$$Z(E/\mathbb{F}_q; T) = \frac{(1 - \alpha T)(1 - \beta T)}{(1 - T)(1 - qT)}$$

where $\alpha, \beta \in \mathbb{C}$ are two complex conjugate numbers with ordinary norm $\sqrt{q}$ in $\mathbb{C}$ and

$$a := \alpha + \beta = \text{Tr}(\phi_\lambda) = 1 + q - \deg(1 - \phi) \in \mathbb{Z}.$$

We can verify that it is the desired form. □

Weierstrass Equation

Lemma

Let (E, O) be an elliptic curve over k . There exists an embedding $\iota : E \hookrightarrow \mathbb{P}^2$ defined over k whose image is the curve defined by a Weierstrass equation

$$Y^2Z + a_1XYZ + a_3YZ^2 = X^3 + a_2X^2Z + a_4XZ^2 + a_6Z^3.$$

Problems

- **Problem 1:** Is the morphism $id - \phi^n$ unramified?

Solution: Use differentials.

- **Problem 2:** Write any elliptic curve in Weierstrass form.

Solution: Use the Riemann-Roch theorem.

- **Problem 3:** Compute the degree.

Solution: Use the Tate module.

Table des matières

- 1 Motivation
- 2 Vocabularies
- 3 Towards the Question
- 4 Differential
- 5 Riemann-Roch
- 6 Tate Module

Kähler Differential

Definition

Let $A \rightarrow B$ be a morphism of rings. We define the B -module $\Omega_{B/A}$ of Kähler differentials as the quotient of the B -module generated by the symbols dx , $x \in B$ by the relations $d(x + y) = dx + dy$, $d(xy) = xdy + ydx$, $dx = 0$ if $x \in A$. It is thus equipped with an A -linear map $x \mapsto dx$. We can also define it as:

$$\Omega_{B/A} := I/I^2, \text{ where } I := \ker(B \otimes_A B \xrightarrow{\text{mult}} B),$$

in which case, $dx = (x \otimes 1 - 1 \otimes x) \bmod I^2$.

We have the universal property: for any B -module M , we have the isomorphism

$$\text{Hom}_B(\Omega_{B/A}, M) \xrightarrow{\cong} \text{Der}_A(B, M),$$

where an A -linear derivation on B is an A -module morphism $d : B \rightarrow M$ satisfying the Leibniz rule $d(fg) = fdg + gdf$.

Properties of Differential Modules

Lemma

We have the following properties for the module of differentials.

- (i) *If B is generated as an A -algebra by elements $x_1, \dots, x_n$, then $\Omega_{B/A}$ is generated by the dx_i as a B -module.*
- (ii) *If $B = A[X_1, \dots, X_n]$, then $\Omega_{B/A}$ is free over B with basis $dX_1, \dots, dX_n$.*
- (iii) *For $A \rightarrow B \rightarrow C$, we have an exact sequence*

$$C \otimes_B \Omega_{B/A} \rightarrow \Omega_{C/A} \rightarrow \Omega_{C/B} \rightarrow 0.$$

- (iv) *For L/K a finite type extension of k , the morphism $L \otimes_K \Omega_{K/k} \rightarrow \Omega_{L/k}$ is an isomorphism if and only if L/K is a separable extension.*

Examples

Example

- (i) Let $L = \mathbb{F}_p(X)$ and $K = \mathbb{F}_p(X^p)$. Since any $\mathbb{F}_p$ -linear derivation of L is zero on K , by the universal property we have $\Omega_{L/K} = \Omega_{L/\mathbb{F}_p} = L \cdot dX$.
- (ii) Let L/K be a separable extension. For any $x \in L$, suppose f is a polynomial that annihilates x such that $f'(x) \neq 0$. Then we have $f(x) = 0$, and hence $0 = df(x) = f'(x)dx$. It follows that $dx = 0$. We conclude that $\Omega_{L/K} = 0$.

Meromorphic Differential on a Smooth Curve

Proposition

Let C be a smooth curve. We denote $\Omega_C := \Omega_{k(C)/k}$, called the space of meromorphic differentials on C .

- (i) Ω_C is a $k(C)$ -vector space of dimension 1.
- (ii) Let $\varphi : C \rightarrow C'$ be a non-constant morphism of elliptic curves. The extension $\varphi^* : k(C') \rightarrow k(C)$ induces a $k(C')$ -linear map

$$\Omega_{C'} \rightarrow \Omega_C, \sum f_i dx_i \mapsto \sum (\varphi^* f_i) d(\varphi^* x_i).$$

This map is non-zero (and thus injective) if and only if φ is separable.

Démonstration.

(Proof of (ii)) We use (iii) and (iv) of the previous lemma. □

Problem 1

Lemma

Let $\varphi : E_1 \rightarrow E_2$ be a non-constant morphism of varieties between two elliptic curves. If φ is non-constant, we have:

(i) $\deg_i(\varphi) = e_P(\varphi)$ does not depend on P .

(ii) $\deg_s(\varphi) = \#\varphi^{-1}(Q)$ for all $Q \in E_2$.

The Riemann-Hurwitz formula can be used to prove this.

Problem 1: Choose a non-zero differential $\omega \in \Omega_{C'}$. Since the Frobenius morphism ϕ is inseparable, we have $(id - \phi)^*\omega = \omega \neq 0$.

Table des matières

- 1 Motivation
- 2 Vocabularies
- 3 Towards the Question
- 4 Differential
- 5 Riemann-Roch
- 6 Tate Module

Divisor

Definition

Let C be a smooth curve. We denote $\text{Div}(C)$ the free abelian group generated by the points of C . If $D = \sum_P a_P [P]$ is a divisor, its degree $\deg(D)$ is the integer $\sum_P a_P$. Let D, D' be two divisors, we say $D \leq D'$ if for all points $P \in C$, the coefficient of D is less than or equal to that of D' . We say D is effective if $D \geq 0$. Furthermore, we define a map

$$\text{div} : k(C)^\times \rightarrow \text{Div}(C), f \mapsto \sum_P v_P(f)[P].$$

Let $\ell(D)$ be the dimension of the k -vector space

$$\mathcal{L}(D) := \{f \in k(C) : \text{div}(f) + D \geq 0\}.$$

Example

Let C be a projective curve. If there exists $P \in C$ such that $\ell([P]) = 2$, then $C \simeq \mathbb{P}^1$.

Riemann-Roch Theorem

Theorem

Let C be a (smooth projective) curve. Then for any $D \in \text{Div}(C)$, we have

$$\ell(D) - \ell(K_C - D) = \deg(D) + 1 - g(C).$$

Note that $\deg(K_C) = 0$ for an elliptic curve. Using Riemann-Roch, we obtain the relationship between the number $\ell(D)$ of a divisor and its degree $\deg(D)$ as shown in the table below.

$\deg(D):$	$\dots$	-2	-1	0	1	2	3	$\dots$
$\ell(D):$	$\dots$	0	0	$0 \text{ or } 1$	1	2	3	$\dots$

That is, $\ell(D)$ is determined by $\deg(D)$ except when $\deg(D) = 0$. But when $\deg(D) = 0$, $\ell(D) = 1$ if D is principal, and $\ell(D) = 0$ otherwise.

Weierstrass Equation

Démonstration.

Since O is a k -rational point, the divisor $3[O]$ is defined over k . By the corollary of Riemann-Roch, we have $\ell(2[O]) = 2$, $\ell(3[O]) = 3$. We can find $x \in \mathcal{L}(2[O]) \setminus \mathcal{L}([O])$, and $y \in \mathcal{L}(3[O]) \setminus \mathcal{L}(2[O])$. Then $\{1, x, y\}$ forms a k -basis of $\mathcal{L}(3[O])$ which provides an embedding $\iota : E \rightarrow \mathbb{P}^2$. To compute an equation, note that the family of 7 functions $\{y^2, x^3, yx, x^2, y, x, 1\}$ lives in $\mathcal{L}(6[O])$ whose dimension is 6. It follows that the family is k -linearly dependent. Moreover, note that the orders of pole at O of these functions are respectively 6, 6, 5, 4, 3, 2, 0, the families obtained by removing y^2 or x^3 are independent. Therefore, both terms must appear in the equation. After a reasonable multiplication, we can arrange the equation in the form where the coefficients of y^2 and x^3 are 1. Thus, $\iota(E)$ is included in a Weierstrass equation C . □

Table des matières

- 1 Motivation
- 2 Vocabularies
- 3 Towards the Question
- 4 Differential
- 5 Riemann-Roch
- 6 Tate Module

Torsion Subgroup

Definition

Let E be an elliptic curve and $m \in \mathbb{Z}^\times$. We define the m -torsion subgroup

$$E[m] := \{P \in E : [m]P = 0\}.$$

Proposition

Let E be an elliptic curve and $m \in \mathbb{Z}^\times$ such that $\gcd(m, \text{char}(k)) = 1$. Then we have

$$E[m] = \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}.$$

Tate Module

Definition (Tate Module)

Let E be an elliptic curve and λ a prime number. We have a projective system

$$\cdots \rightarrow E[\lambda^{n+1}] \xrightarrow{[\lambda]} E[\lambda^n] \rightarrow \cdots \rightarrow E[\lambda].$$

The (λ -adic) Tate module is the projective limit of this system

$$T_\lambda(E) := \varprojlim_n E[\lambda^n]$$

which is naturally a $\mathbb{Z}_\lambda$ -module.

Structure of the Tate Module

Proposition

If $\lambda \neq \text{char}(k)$, we have

$$T_\lambda(E) \simeq \mathbb{Z}_\lambda \times \mathbb{Z}_\lambda.$$

Démonstration.

We take the projective limit in the previous proposition. □

Isogenies and Tate Modules

Let $\varphi : E_1 \rightarrow E_2$ be an isogeny between two elliptic curves, then φ induces a map $E_1[\lambda^n] \rightarrow E_2[\lambda^n]$, hence a $(\mathbb{Z}_\lambda\text{-linear})$ map between Tate modules

$$T_\lambda(E_1) \rightarrow T_\lambda(E_2).$$

We obtain a map

$$\mathrm{Hom}(E_1, E_2) \rightarrow \mathrm{Hom}(T_\lambda(E_1), T_\lambda(E_2))$$

and in particular for any elliptic curve,

$$\mathrm{End}(E) \rightarrow \mathrm{End}(T_\lambda(E)), \psi \mapsto \psi_\lambda$$

which is indeed a ring homomorphism. After choosing a $\mathbb{Z}_\lambda$ -basis for $T_\lambda(E)$, we can write ψ_λ as a 2×2 matrix in $GL_2(\mathbb{Q}_\lambda)$.

Completion of the Proof

Proposition

Let E be an elliptic curve, $\psi \in \text{End}(E)$ an endomorphism. Recall that we have a Tate module endomorphism $\psi_\lambda \in \text{End}(T_\lambda(E))$. Then we have

$$\det(\psi_\lambda) = \deg(\psi)$$

and

$$\text{Tr}(\psi_\lambda) = 1 + \deg(\psi) - \deg(1 - \psi).$$

In particular, $\det(\psi_\lambda)$ and $\text{Tr}(\psi_\lambda)$ do not depend on the choice of λ .

Completion of the proof. (On blackboard)

An example. (On blackboard)

Bibliography

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